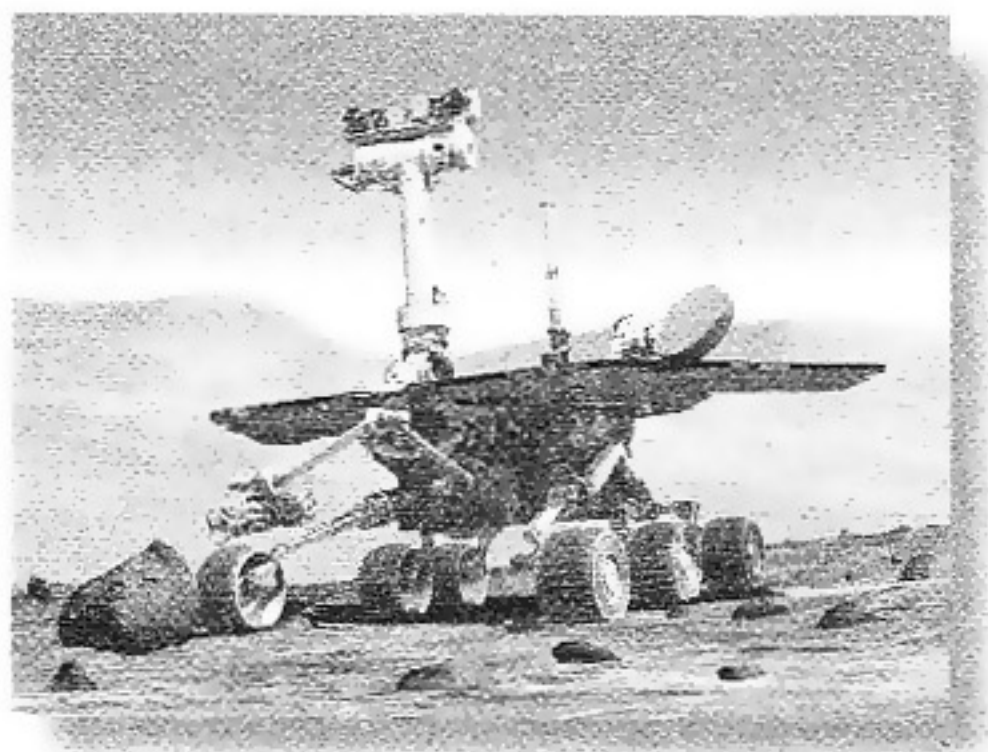


13 SCIENCE AND TECHNOLOGY



The science of today is the technology of tomorrow.

Edward Teller

Modern technology

Owes ecology

An apology.

Alan M. Eddison

□ VOCABULARY INPUT

Dealing with this topic you will come across the following vocabulary. Read it and look up the unknown words in your dictionary.

- **science:** scientist, researcher, inventor, explorer
to conduct research, to experiment, to study, to observe, to invent, to discover, to explore, to search for, to achieve, to apply
- **technology:** development, technological change, machine manufacturing, steam power, powered machinery
- **inventions:** wheel, gun powder, windmills, watermills, steam engine, printing machine, electric bulb, telegraph, telephone, laser, nuclear power
- **technological progress:** transport, medicine, biology, information and communication technology
- **everyday usage of modern technology:** household appliances, gadgets, devices, machines
- **pitfalls of using modern technology in life:** abuse, misuse, threat, pollution, damage, hurt, growth of consumer society, materialism, social/psychological/spiritual hunger, erosion of privacy and anonymity, brain drain

□ DISCUSSION QUESTIONS

Before reading the text, discuss the following questions in class.

1. Which inventions from the past are essential for mankind?
2. What would our life be like without new inventions and technologies?
3. Can you think of any ways that science can be misused? Explain.
4. Why is human cloning still an issue that is discussed among researchers?
5. How can the Internet be misused?
6. What type of mobile phone will your children be using one day?
7. What are the biggest problems associated with communicating via mobile phone or via the Internet?
8. Name some examples of pollution caused by technology.
9. How is your life affected by living in a consumer society?
10. Can you name any famous scientists? What are they famous for?

□ TEXT

Read the texts below in silence. Check the meaning of the words in bold and answer the questions below.

SCIENCE AND TECHNOLOGY

Science is the systemized knowledge **gained** through experimentation, **observation** and study. There exist many **fields** of sciences – natural sciences, social sciences, environmental sciences and so on. Many **discoveries** (such as the fact that the Earth is not flat) were made **through** observation. Some discoveries lead to the **invention** of an object, a process or a technique. Technology (a product of science) is the application of knowledge **to achieve** a practical result. Society depends on technology and technology exists thanks to the needs of society. Technological development changes society. However, our world has not always been like that. Inventions from the past have contributed enormously to technological development. For example, the invention of the **wheel** had an **incalculable** impact on transportation. **Watermills** and **windmills** were perfect energy **providers**. The industrial revolution, which started in the 18th century in England, brought **major** technological change. The economy based on **manual labour** was replaced by an economy dominated by machine manufacturing. Major **achievements** were the invention of **steam power** and **powered machinery**. Since then, technology has become a part of our everyday life.

1. What is “science” and “technology”? How are they related?
2. Which inventions from the past are mentioned in the passage above? Why are they crucial to our lives?

PROGRESS IN TRANSPORT, MEDICINE AND BIOLOGY

Society has made **enormous scientific** and technological **progress** in transport, medicine, biology and information technology. We realise this when we compare how people travelled in the 18th century and how they travel now. Progress in the car industry and in **aviation** is really **incredible**. In the middle of the 20th century people were able to leave the planet for the first time and **explore space**. *Yuri Gagarin*, the Soviet cosmonaut, was the first human who entered **outer space**. The American astronaut, *Neil Armstrong*, who died in 2012, was the first man to set foot on the Moon.

Great progress has been made in the field of medicine. Science and technology have helped **provide pain relief** for certain types of diseases. It has even **cured** and **prevented** some **diseases** that were considered **incurable** in the past. It helps **prolong** people's lives and makes the lives of **the handicapped** easier. In the field of biology, especially microbiology, scientists do not prolong life, but create it. “**In vitro**” **fertilisation** and **cloning** are currently **the most debated issues** in this field. The first one presents a technique for the **treatment** of **infertility** in which **egg cells** are **fertilised** outside the woman's body. Cloning an organism means creating a new organism with the same

genetic information as a **cell** from an existing one. This invention in molecular biology can be used for many **purposes**, e. g. to produce organs or **tissues** to repair or **replace damaged** ones. However, many people do not agree with human cloning saying that the **potential for abuse** is very big.

3. How has technology made our lives easier with regard to travel?
4. Can you name any examples of scientific research in the field of medicine?

INFORMATION AND COMMUNICATION TECHNOLOGY

During the last few **decades**, we have **witnessed** a huge **advance** in information and communication technology, especially in **wireless** technology. The most **remarkable** advances were the inventions of the **mobile phone** and the Internet. At the beginning of the 20th century, nobody would have been able to imagine what life would be like at the beginning of the 21st century. Technology **evolves** very quickly and the process is unstoppable. Japan and the USA compete to be the first to introduce the latest “**high-tech**” product into the market. The amount of people using mobile phones has **rapidly increased**, especially in Central and Eastern Europe. Mobile phones have become either a symbol of comfortable, quick communication or a fashion, mainly among **teenagers**. Newer and newer applications for mobile phones are being developed; they, for example, allow users to download CD quality songs to their wireless phones. The mobile phone, the music player, the camera and the **palm computer** are being put together into one little **gadget**. **Current trends** show that gadgets are getting more and more **sophisticated** and young people (even children) are able to use them without difficulty. A computer **skill** is becoming a basic skill **acquired** early in childhood. Information and communication technologies are also getting more and more **accessible**. Computers have already replaced people in some professions.

5. Based on the passage, summarise the ideas about the latest high-tech inventions and products.
6. What impact on society is there when people are replaced by computers?

PROS AND CONS OF NEW INVENTIONS

Science and technology help us with **minor** everyday work (**household appliances**) but also with important global issues like saving human lives or exploring space. Without it our lives would be much more difficult and our knowledge restricted. However, science and technology can be **misused**, as has already happened many times. Nuclear power is used in **nuclear power plants** which **generate** electricity. Many organisations, such as the independent, campaigning organisation *Greenpeace*, **promote** alternative ways of producing electricity. The safety of nuclear power plants is **debatable**. They **proved** to be dangerous in the 1980s in Chernobyl (the Ukraine), where a massive nuclear explosion put the rest of Europe in danger, and in 2011 in Fukushima in Japan. The inventions of nuclear physics are also used in medicine, both in **diagnosis** and therapy and in other fields. How-

ever, the **armaments industry**, especially in producing **nuclear weapons**, has become the most dangerous **threat** to the world we live in.

7. Which pros and cons of new inventions are discussed in the text above?
8. In which areas of science are the inventions of nuclear physics used?

WHAT'S OUR FUTURE?

Technology has created a **consumer lifestyle**, the opposite extreme of **poverty**. The consumer society has been very **harmful** to the environment and it does not provide us with a **sense of fulfilment**. It forces us to **rely on** material things because we **suffer from** social, psychological and **spiritual hungers**. It also causes physical damage because this type of lifestyle causes many diseases. Scientists will continue to **conduct** research in medicine, microbiology and all the other fields of science. Slovakia has many good researchers, but in recent years a lot of highly qualified scientists, programmers, technicians, doctors and teachers have left the country to find a better paid job elsewhere. This **phenomenon**, called the **brain drain**, presents a serious **loss** for our country and is caused by a **lack of** job opportunities and low salaries. It is therefore very important to improve the economy **as a whole** and create good working **conditions**, so that this trend does not continue for much longer.

9. Do you agree with the fact that today's consumer lifestyle is the by-product of technology? Explain.
10. Which factors cause a brain drain?

□ VOCABULARY LIST

abuse [ə'bjʊ:s] – zneužitie
accessible [ək'sesəbl] – dostupný
to achieve – dosiahnuť
achievement – úspech
to acquire [ə'kwaiə(r)] – osvojiť si
advance [əd'va:ns, AmE əd'væns] – pokrok
armaments industry – zbrojný priemysel
aviation [eɪvɪ'eɪʃn] – letectvo
brain drain – odliv mozgov
cell [sel] – bunka
cloning – klonovanie
condition – podmienka
to conduct – viesť
consumer lifestyle – konzumný životný štýl
to cure [kjʊə] – vyliečiť
current trends – súčasné trendy
damaged – poškodený
debatable [dɪbeɪtəbl] – sporný, diskutabilný
decade – dekáda
diagnosis [daɪəg'nəʊsɪs] – diagnostika
disease – choroba
discovery – objav
egg cell – vajíčko (v tele ženy)
enormous – obrovský
to evolve – vyvíjať sa
to fertilise (AmE: fertilize) [fɜ:təlaɪz, AmE: fɜ:rtəlaɪz] – oplodniť
field – oblasť
gadget [gædʒɪt] – prístroj
to gain – získať
to generate – vyrábať
genetic [dʒə'netɪk] – genetický
harmful – škodlivý
high-tech – na vysokej technickej úrovni
household appliance [haʊshəʊld ə'plaɪəns] – domáci spotrebič
incalculable impact – nepredstaviteľný dopad
incredible – neuveriteľný
incurable – nevyliečiteľný
infertility [ɪnfə'tɪlɪtɪ] – neplodnosť
invention – vynález
in vitro fertilization (also fertilization) [ɪn vi:trəʊ fɜ:təlaɪ'zeɪʃn] – oplodnenie v skúmavke
lack (of) – nedostatok
loss – strata
major [meɪdʒə(r)] – hlavný
manual labour (AmE: labor) – manuálna práca
minor [maɪnə(r)] – drobný
to misuse – zneužiť

mobile phone (AmE: cell phone) – mobilný telefón
nuclear power plant – jadrová elektráreň
nuclear weapons [nju:kliə(r) wepəns] – jadrové zbrane
observation – pozorovanie
outer space – vesmír
palm computer – počítač do ruky
phenomenon [fə'nɒmɪnən] (pl. phenomena) – jav
potential [pə'tenʃl] – potenciál
poverty – chudoba
powered machinery – stroje poháňané elektrickou energiou
to prevent – predchádzať
to prolong – predĺžiť
to promote – propagovať, presadzovať
provider [prə'vaɪdə(r)] – poskytovateľ
to prove [pru:v] – dokázať
purpose – účel
to realise (AmE: realize) [rɪəlaɪz] – uvedomiť si
to rely on – spoliehať sa na
remarkable – pozoruhodný
to replace – nahradiť
science [saɪəns] – veda
scientific progress [saɪəntɪfɪk] – vedecký pokrok
skill – zručnosť
sophisticated – zložitý, vyspelý
steam power [sti:m paʊə(r)] – parný stroj
to suffer from – trpieť na
threat [θret] – hrozba
the handicapped – postihnutí
through [θru:] – cez
tissue [tɪʃu:, also tɪʃju:] – tkanivo
treatment – liečba
watermill – mlynské koleso
wheel [wi:l] – koleso
windmill – veterný mlyn
wireless – bezdrôtový
to witness – byť svedkom

as a whole [həʊl] – ako celok
to explore space – preskúmať vesmír
to increase rapidly – prudko zvýšiť
the most debated issues – najdiskutovanejšie témy
to provide pain relief – poskytnúť úľavu od bolesti
sense of fulfilment – pocit naplnenia
spiritual hunger – duševný hlad

□ VOCABULARY PRACTICE

Task 1 From the list below, choose the inventions/machines/gadgets/household appliances/means of transport which you use daily and also the things which you do not use at all. Then, decide which thing is the most important and the least important for you and say why.

hair dryer – washing machine – mobile phone – car – lift – fridge – kettle – microwave oven – television – computer – cooker – lighter – television – CD player – digital camera – toaster – vacuum cleaner – dishwasher – coffee machine – laptop – MP3 player – alarm clock – tablet – play station – e-book reader – drill – satellite dish – fan – printer – Xerox machine – fax machine – answering machine – bicycle – air-conditioner – palm computer – watch – air-humidifier – calculator – digital photo frame

I daily use: _____

I never use: _____

The most important for me is: _____

The least important for me is: _____

Task 2 What do children/teenagers/adults use computers for? Choose from the box.

write documents – scan pictures – play games – browse the Internet – watch DVDs – listen to CDs – make music – download music – read on-line books – learn English – send e-mails – do on-line shopping – draw pictures – save photographs – burn CDs – prepare presentations – take notes – chat on the Internet – pay bills – look for partners – control household heating – watch films – read blogs – use Facebook – make videos – read news – Skype/ICQ

children: _____

teenagers: _____

adults: _____